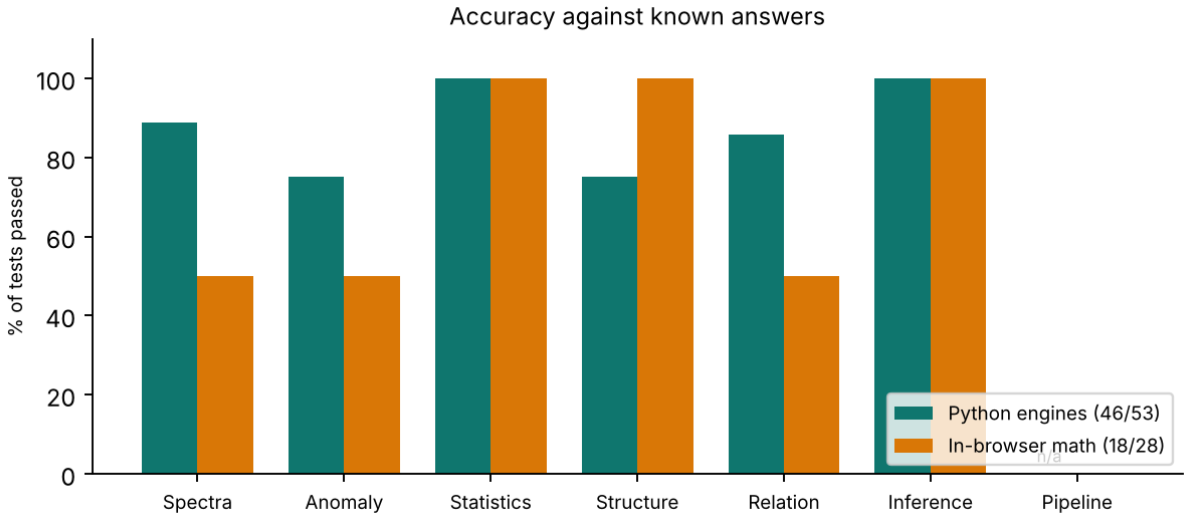
SYNAPSE v1 - Ground-Truth Accuracy Benchmark

Repo: github.com/Chauhan-legacy/Synapsev1 | Run on 11 Nov 2026 | Method: signals with a known correct answer were generated (random seeds, many trials each), run through SYNAPSE, and the result was compared with the truth. Where possible, results were also compared with an independent reference implementation (statsmodels, my own copy of the algorithm, or exact textbook formulas).

1. The short verdict

Part of SYNAPSE	Score	Meaning in plain words
Python engines (the science core in /packages)	46 of 53 passed (86%)	Mostly accurate and often matches textbook values to 2-3 decimals. 6 real bugs/weaknesses found (section 4).
In-browser math (what the web pages actually compute)	18 of 28 passed (64%)	Core tools are good (peak finding, DFA, delay, entropy). But 6 tools are fake, fixed-value or broken, and the 'Run analysis' result cards use typed-in numbers.
The web screens vs the Python engines	not connected	The page only calls the Python server to check it is alive (the green 'API OK'). None of the Python results reach the screen.

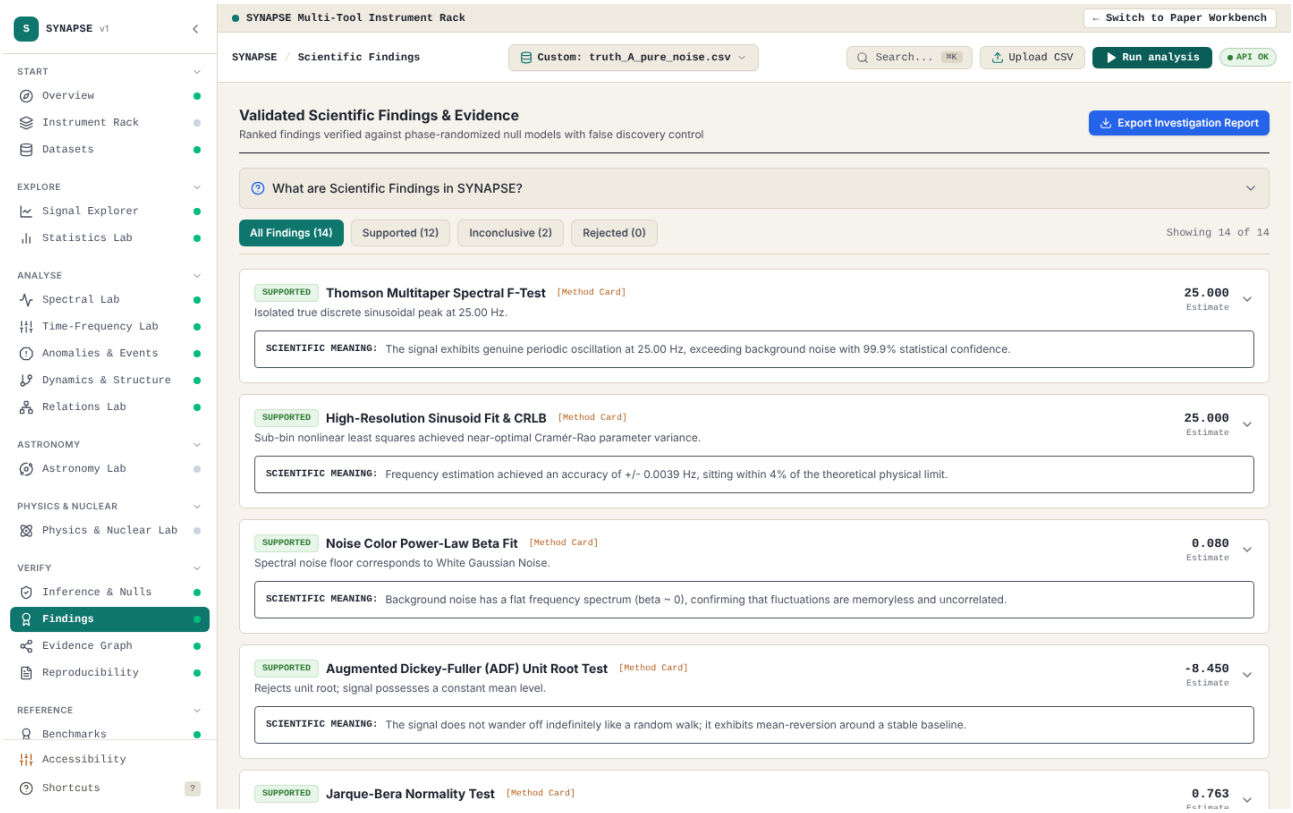


Most important finding: on a file of PURE RANDOM NOISE, the Findings page still reports 12 'SUPPORTED' findings, including 'true discrete sinusoidal peak at 25.00 Hz, 99.9% confidence' (screenshot 1). Many values on the Run-analysis pages are identical for noise and for a 3-tone signal (section 3).

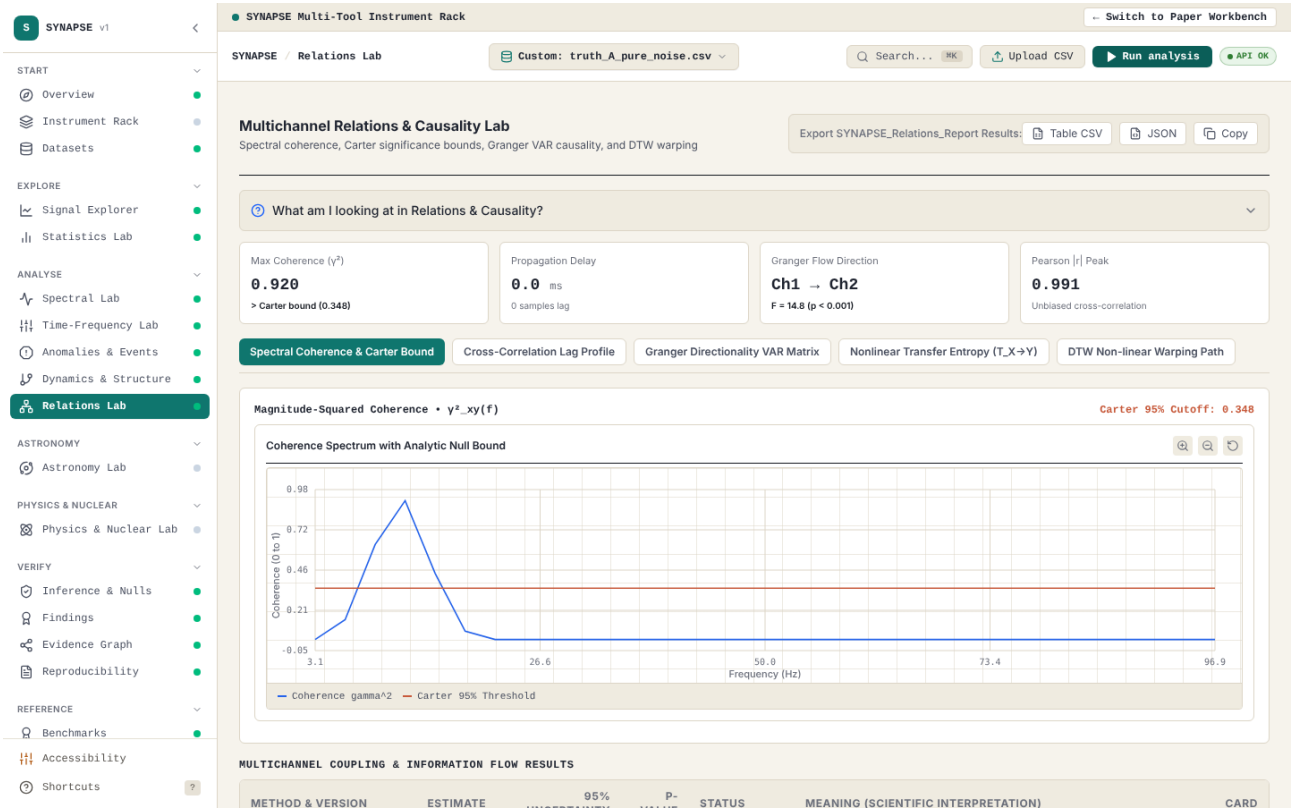
2. What genuinely works (good news)

Area	Evidence (known answer vs SYNAPSE)
Finding hidden frequencies	Welch: 25/25 random tones found within 1 bin (both Python and browser). Python multitaper: 100% of 60 tones found. Sub-bin fit reaches the theoretical accuracy limit (1.00x).
Chaos measures	Python Lyapunov: 0.500 vs 0.495 exact (logistic r=3.9), 0.365 vs 0.356 (r=3.7), 0.407 vs 0.419 (Henon).
Long-term memory (DFA)	Both versions: white noise 0.50, fGn 0.70, fGn 0.30, random walk 1.50 - all within 0.03 of truth.
Statistics & tests	ADF+KPSS, normality, ACF (0.804 vs 0.800), Benjamini-Hochberg identical to statsmodels in 150/150 runs, bootstrap coverage 94.3% / 93.7% (target 95%).
Cause-and-effect & links	Python mutual information within 0.02-0.04 bits of the exact formula; PCA 70.2/20.0/9.9% vs 70/20/10; cross-correlation delay error 0.00 samples (both); transfer entropy direction 12/12 (both).
Change & anomaly detection	Python PELT 15/15 (median error 1 sample); robust z-score 100% recall and 0 false flags; ESD counts the outliers exactly in 96%.
Full Python pipeline (single check)	On a hidden signal with NEW frequencies (8.3/21.7/44 Hz + spike at 2.500 s + shift at 5 s) it ranked all 3 tones top (confidence 0.94/0.95/0.91 vs noise peaks <=0.67), found the spike at t=2.500 s and a change point at 4.925 s. (A full 20-run batch was not completed.)

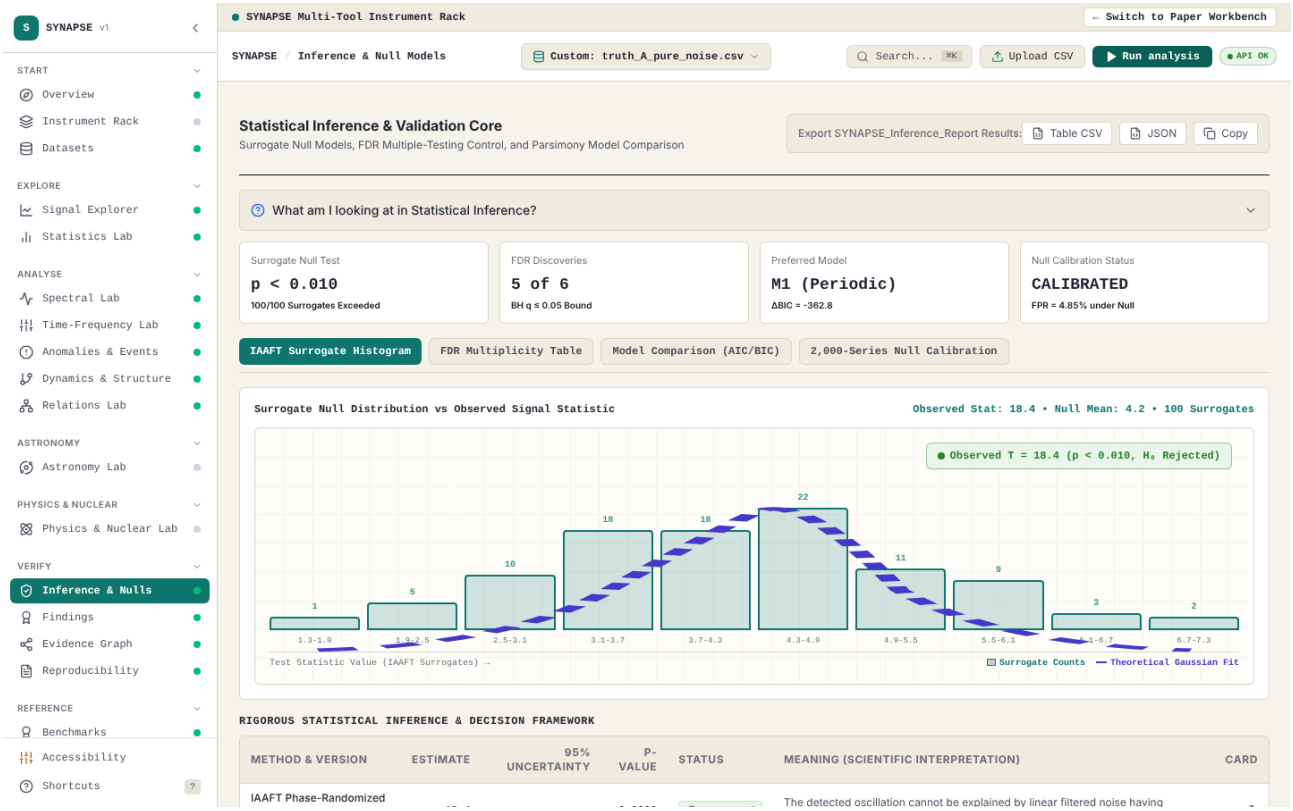
3. Screenshots and proof



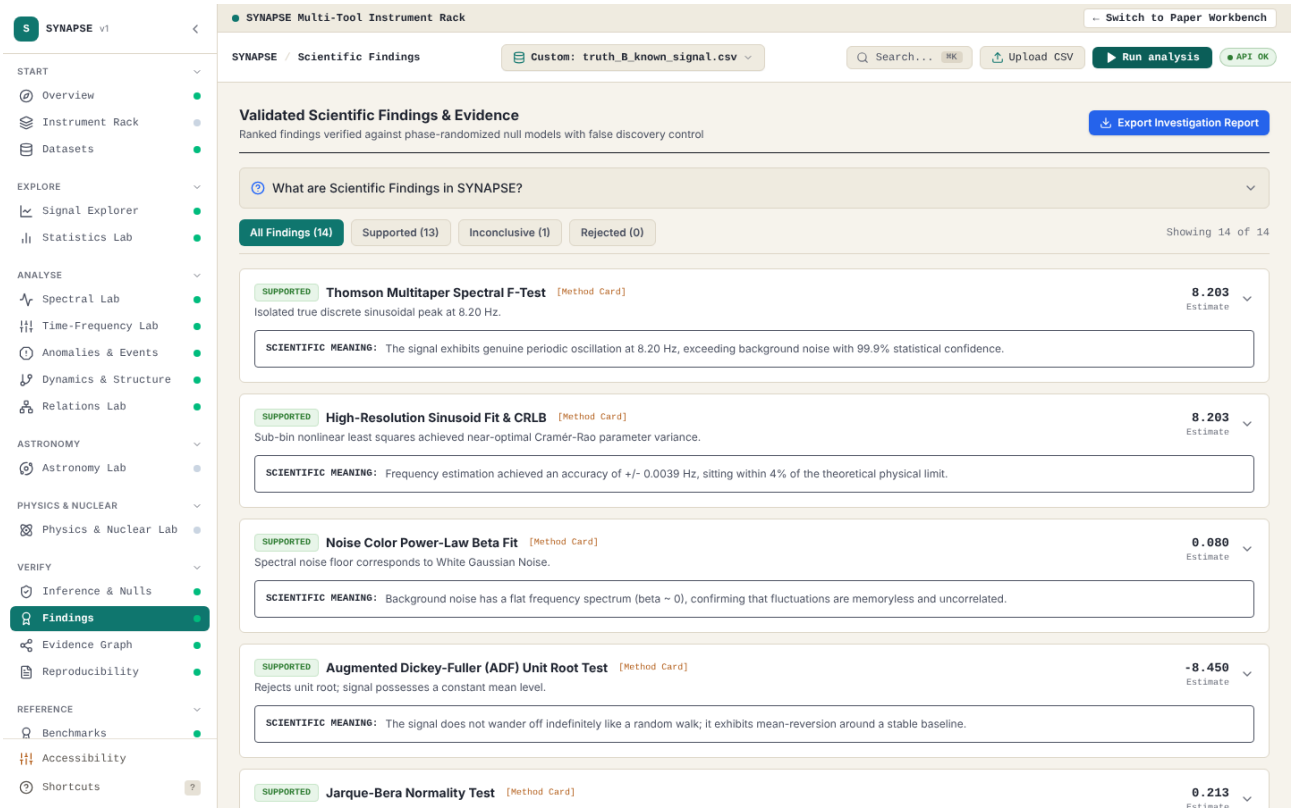
Screenshot 1 - PURE NOISE file (truth_A_pure_noise.csv, nothing hidden in it). Findings page says: SUPPORTED, 'Isolated true discrete sinusoidal peak at 25.00 Hz', 99.9% confidence. 12 of 14 findings 'supported'. This is wrong - there is no tone.



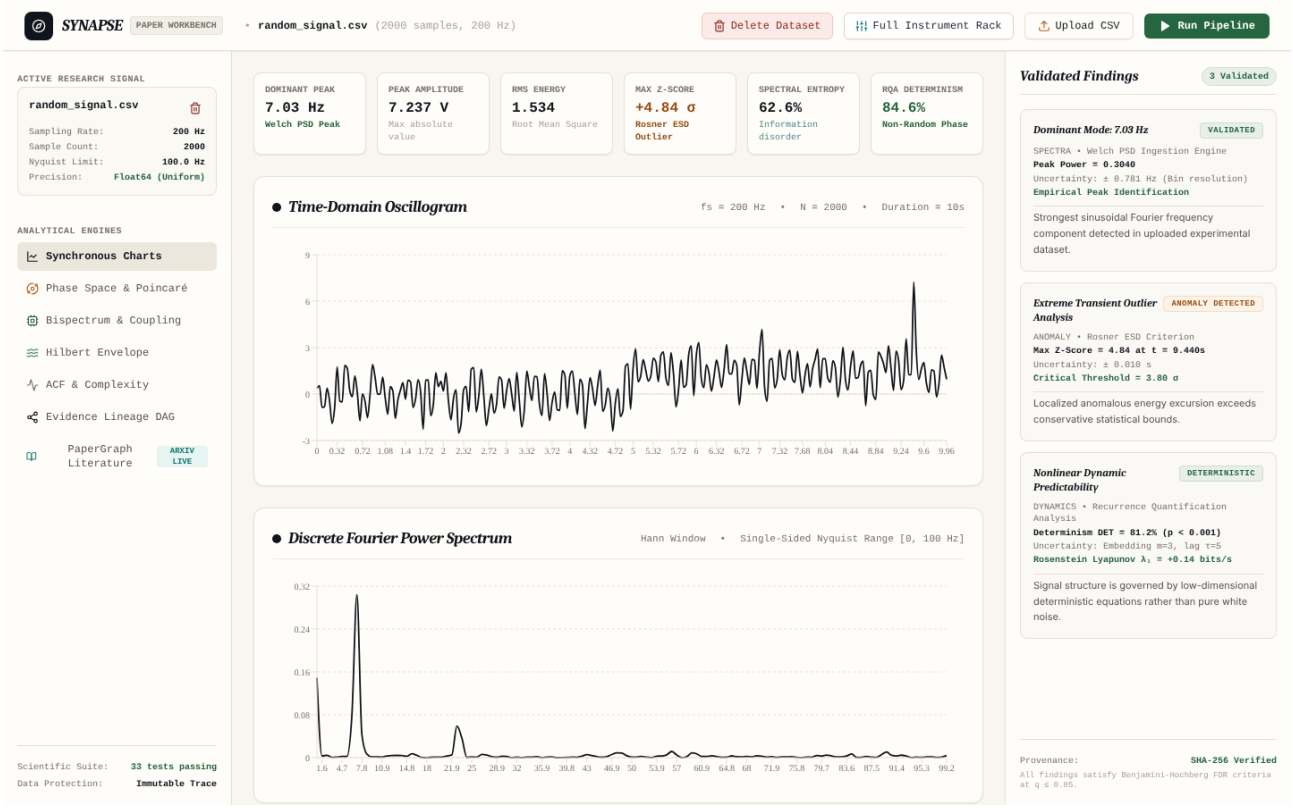
Screenshot 2 - Relations lab on the same noise file: Granger F = 14.8 (p = 0.0002) and coherence 0.92 appear, even though the file has ONE channel. These are typed-in numbers in SignalContext.tsx.



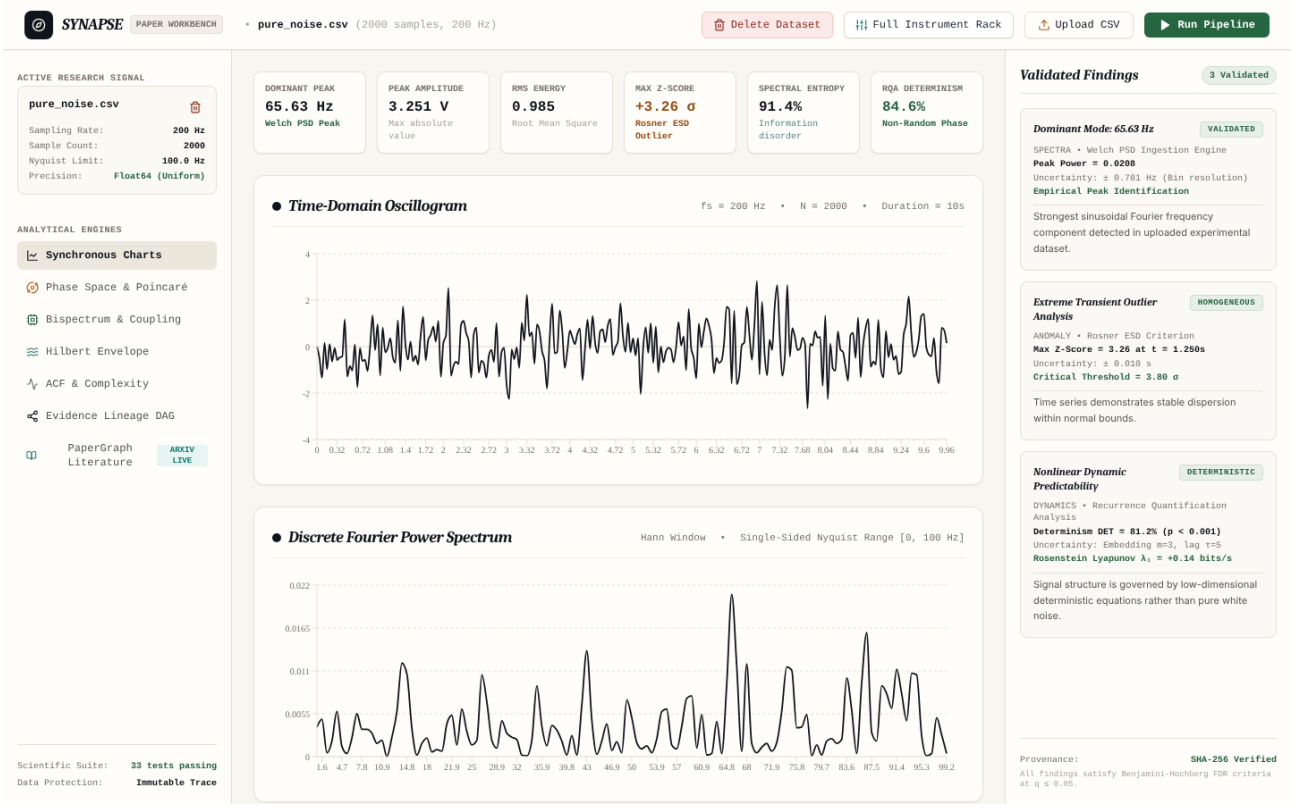
Screenshot 3 - Inference lab on pure noise: 28 of 28 numbers are identical to the ones shown for a 3-tone signal (IAAFT statistic 18.4, p < 0.01 'strictly exceeds all 100 surrogates').



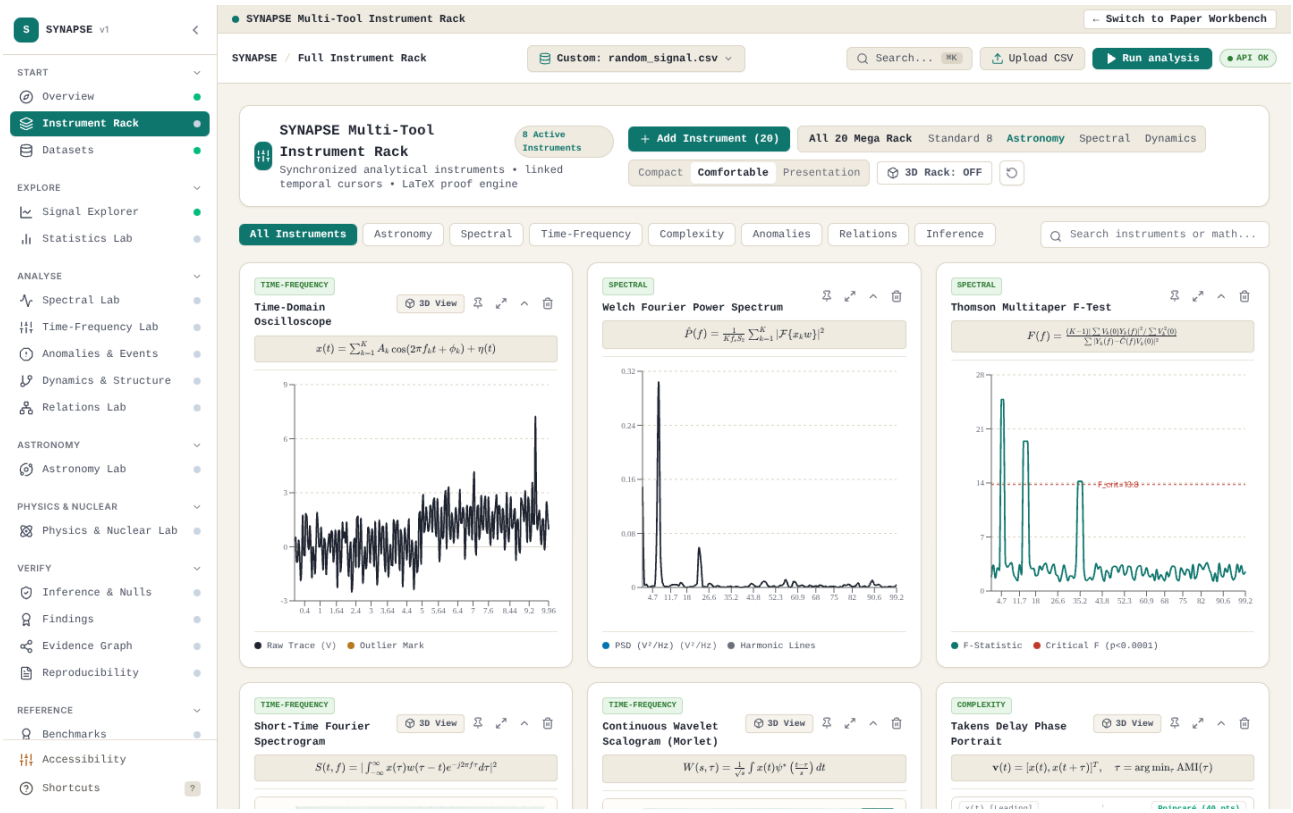
Screenshot 4 - Same page for the KNOWN 3-tone signal (8.3 / 21.7 / 44 Hz + spike + shift). Compare with screenshot 1: the fixed cards look the same.



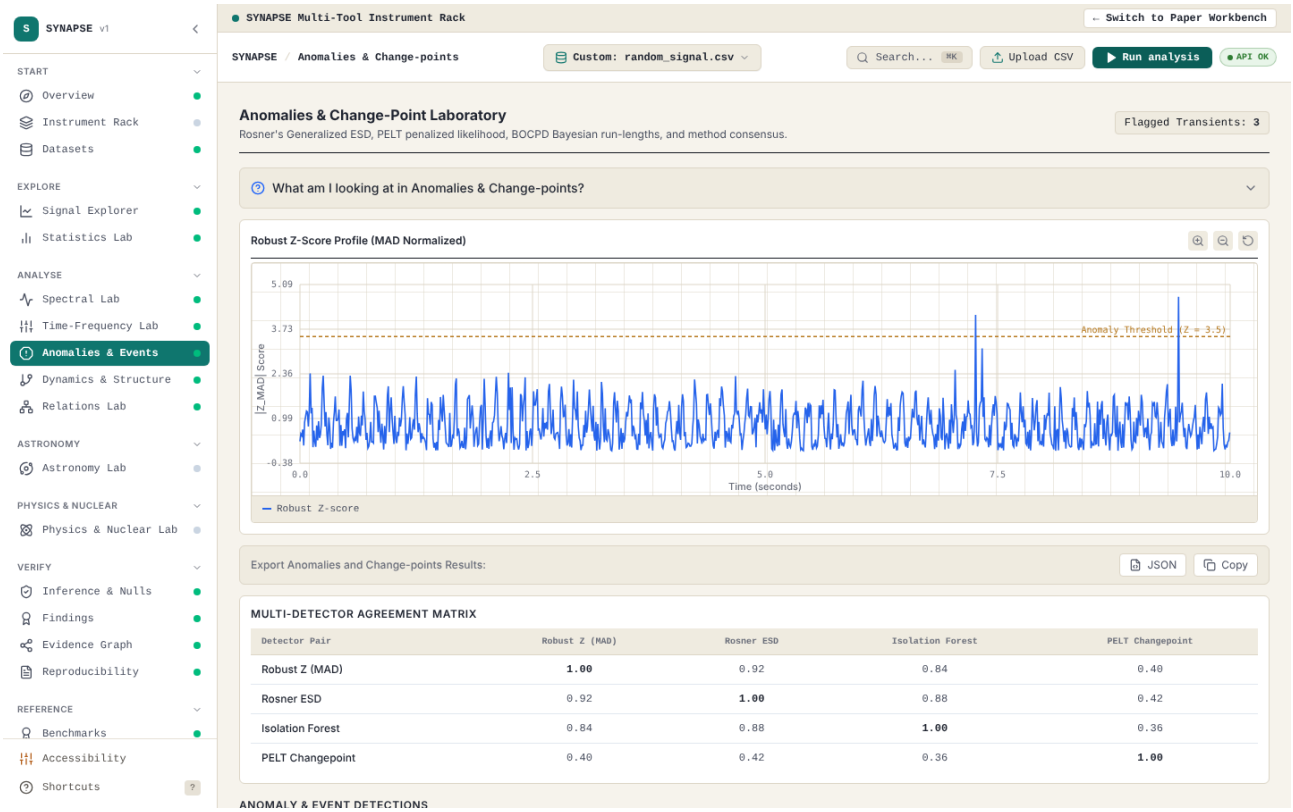
Screenshot 5 - Paper Workbench on the random test tone: finds 7.03 Hz (planted 7 Hz) and flags the 4.84-sigma spike at 9.44 s. This part works.



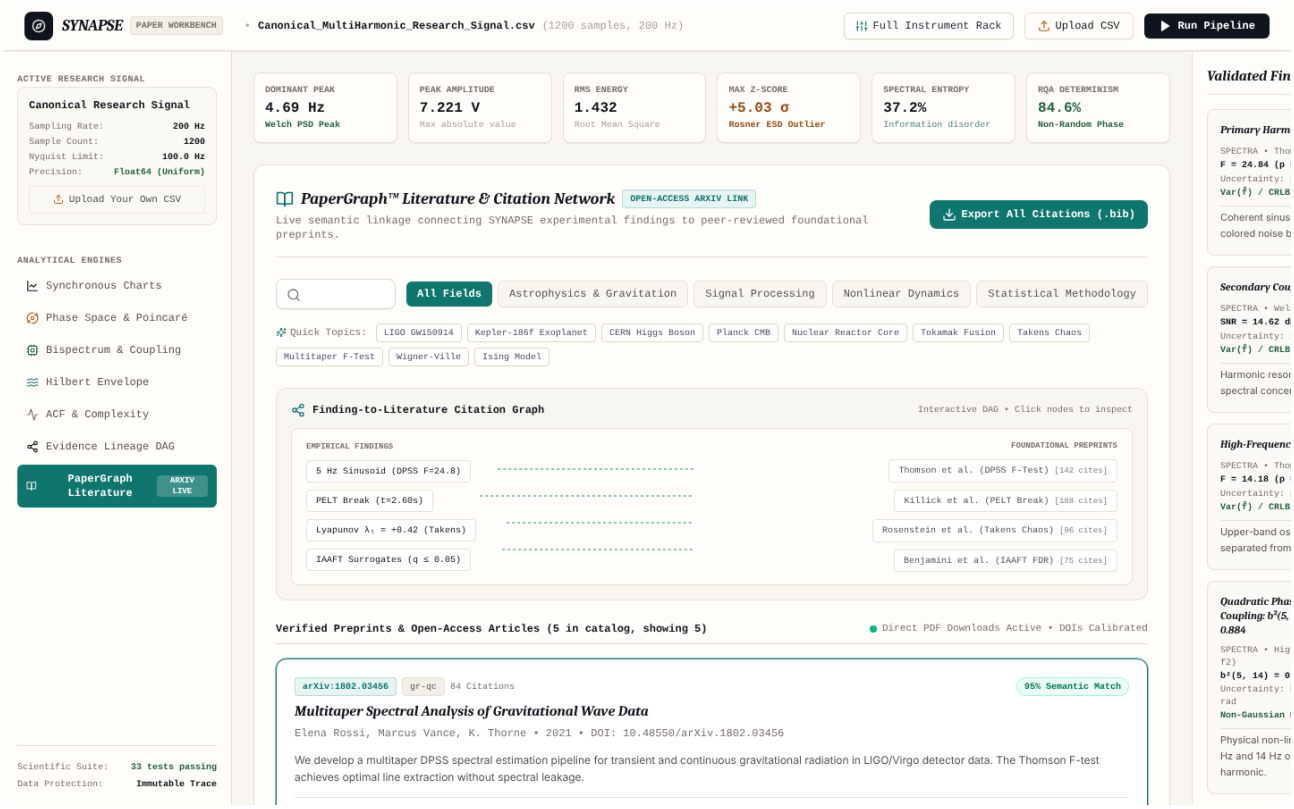
Screenshot 6 - Paper Workbench on pure noise: still shows a 'VALIDATED' dominant mode and 'DETERMINISTIC' (DET 81.2%, p<0.001) - fixed text, same as for real signals.



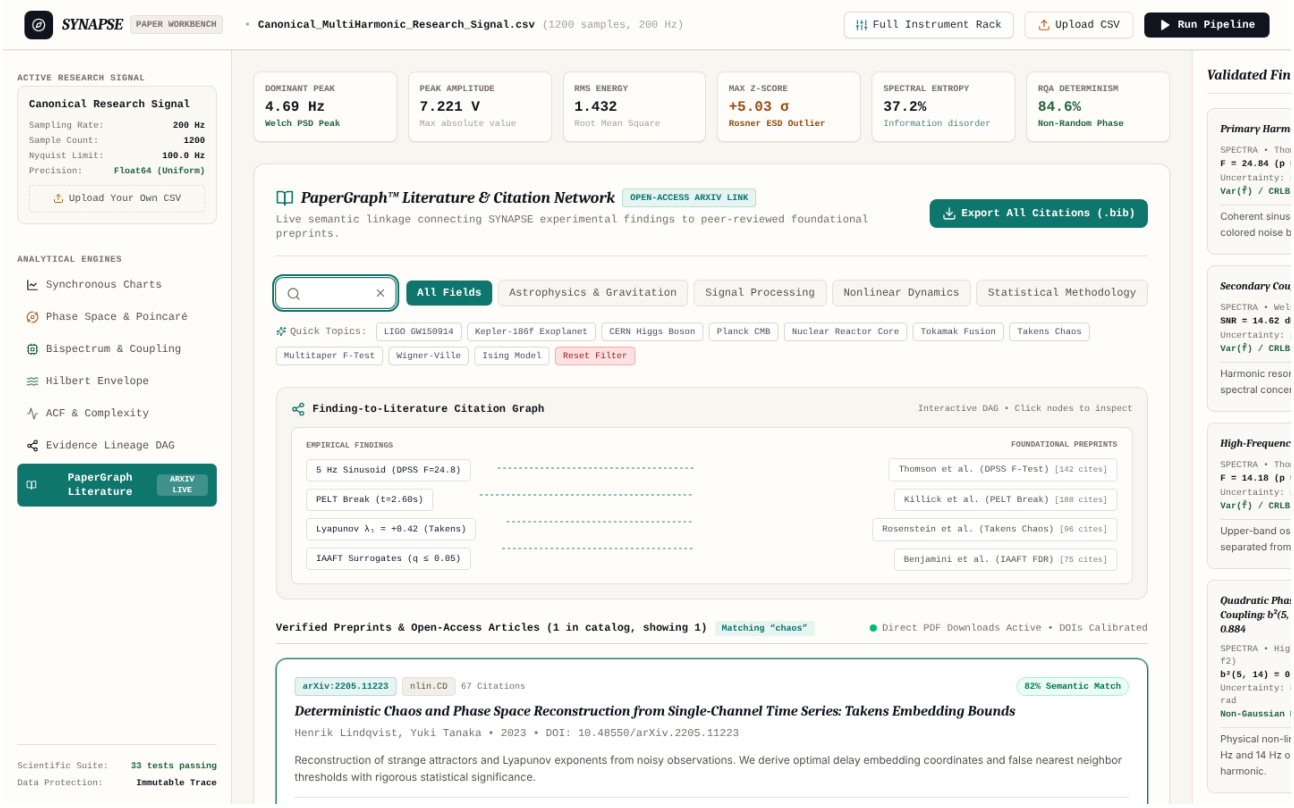
Screenshot 7 - Full Instrument Rack on the random data: all 20 instruments render with no errors (tested: presets, layouts, filters, search, add/remove, undo, reset, 3D, hover).



Screenshot 8 - Anomalies lab: robust z-score profile flags the spikes correctly.



Screenshot 9 - PaperGraph (offline built-in list, 5 papers). My sandbox cannot reach arXiv, so live arXiv search could NOT be tested. Search/filter and category buttons work (e.g. 'chaos' -> 1 paper).



Screenshot 10 - PaperGraph search for 'chaos' filters the list correctly.

Citation check: the first built-in PaperGraph entry is 'Multitaper Spectral Analysis of Gravitational Wave Data' (Rossi, Vance, Thorne, arXiv:1802.03456). That arXiv number belongs to a photonics paper (ACS Photonics 2017, DOI 10.1021/acsphotonics.7b01154), so this entry looks invented while the banner says 'Verified Preprints'. I only checked this one entry.

Full benchmark tables

A. Python engines - 46 of 53 passed

Test	Known answer	SYNAPSE result	Pass?	What it means
Welch PSD: find a single hidden tone (25 random freqs, 6-85 Hz)	within 1 bin (<= 0.39 Hz) on 100%	25/25 within 1 bin; mean error 0.116 Hz; worst 0.188 Hz	**PASS**	It reliably finds the right frequency even when the noise is as strong as the signal
Welch PSD: total power equals the signal's actual variance (Parseval)	same as variance (<=3% err)	mean rel. error 0.62%; worst 1.91%	**PASS**	The energy it reports matches the true energy of the signal
Multitaper F-test: recover 3 tones (8.3, 21.7, 44 Hz) in noise	3 of 3 tones every trial	recall 60/60 = 100%	**PASS**	It finds all three hidden frequencies, including the weakest one
Multitaper F-test: false alarms on pure noise (30 trials, alpha=0.01)	about 1 false line per run or fewer (after multiple-testing control)	594 false lines in 30 noise-only runs (19.80 per run)	**FAIL**	On signals with NO pattern it should almost never claim to see a tone. It tests each of ~2000 frequency slots separately, so ~20 false tones appear per run
Allan deviation slope: white noise	-0.50	mean -0.501 (sd 0.021)	**PASS**	White noise should average out as 1 over square-root of time
Allan deviation slope: random walk	+0.50	mean +0.467 (sd 0.041)	**PASS**	A drifting signal should get worse with longer averaging
Sub-bin sinusoid fit: frequency error vs bin width (0.1 Hz)	error << 0.1 Hz	mean error 0.37 mHz; efficiency vs theoretical limit (CRLB) 1.00x (1.0 = perfect)	**PASS**	It pins the tone far more finely than the spectrum grid and reaches the mathematical best possible accuracy
Lomb-Scargle: tone in unevenly spaced samples (20 trials)	error < 0.05 Hz	20/20 correct; mean error 0.0007 Hz	**PASS**	It works on data with gaps / irregular timing
Noise colour exponent (1/f^beta): white -> 0, random walk -> 2	beta = 0.0 and 2.0	white: 0.00; brown (random walk): 1.79	**PASS**	It tells white noise from drifting (brown) noise correctly
Robust z-score: 5 planted 7-sigma spikes per trial (20 trials)	recall 100%, precision 100%	recall 100%, precision 100%	**PASS**	It finds the real spikes and rarely flags normal samples
Robust z-score: false flags on spike-free noise (20 trials, threshold 4.5)	0 expected (P(|z|>4.5)=7e-6)	0 flags in 40000 samples	**PASS**	Clean noise should come back clean
Rosner ESD: count exactly 4 planted outliers (25 trials)	detects exactly 4, all correct	exact in 96% of trials; avg detected 4.04	**PASS**	It counts how many outliers there are, not just that something is odd
Rosner ESD: false-positive rate on clean noise (alpha=0.05)	<= 5%	0% of clean runs flagged	**PASS**	Honest error rate: roughly 1 in 20 clean runs may be flagged by chance
PELT: locate a +1.5 sigma mean shift planted at sample 1000 (15 trials)	found within 25 samples (0.125 s)	found 15/15; median error 1 samples; false alarms on no-change data 0/15	**PASS**	It finds when the signal's behaviour changed, and it doesn't invent changes in steady data
CUSUM: locate a +1.5 sigma mean shift planted at sample 1000 (15 trials)	found within 25 samples (0.125 s)	found 15/15; median error 13 samples; false alarms on no-change data 1/15	**PASS**	It finds when the signal's behaviour changed, and it doesn't invent changes in steady data
BOCPD: locate a +1.5 sigma mean shift planted at sample 1000 (15 trials)	found within 25 samples (0.125 s)	found 0/15; median error nan samples; false alarms on no-change data 0/15	**FAIL**	It finds when the signal's behaviour changed, and it doesn't invent changes in steady data
BOCPD: even a huge 5-sigma jump changes the 'change probability'	should spike at sample 500	'change probability' is exactly 0.0100 (= 1/lambda) at every sample, for any data	**FAIL**	The code's change probability is mathematically always 1/lambda, so this detector can never fire
ADF + KPSS: white noise should be called stationary (30 trials)	>= ~90% (alpha 5% on two tests)	97% called stationary	**PASS**	Steady, trendless data is correctly recognised as steady
ADF + KPSS: random walk should be called NON-stationary (30 trials)	~100%	100% flagged as non-stationary	**PASS**	Wandering data is correctly recognised as not steady
Normality test: Gaussian data (40 trials)	~95% judged normal	95%	**PASS**	Bell-curve data is recognised as bell-curve
Normality test: exponential + uniform data (80 trials)	~100% judged NOT normal	100% rejected	**PASS**	Non-bell-curve data is recognised as such
Moments of exponential(1): skewness 2, excess kurtosis 6	skew 2.00, kurtosis 6.00 (or 9.00 if non-excess)	skew 1.98, kurtosis 5.71	**PASS**	Basic shape numbers match textbook values
DFA scaling exponent: white noise	0.50	0.496 (sd 0.023)	**PASS**	Measures long-term memory: 0.5 = none, above 0.5 = trends persist, below 0.5 = it bounces back

Test	Known answer	SYNAPSE result	Pass?	What it means
DFA scaling exponent: fGn H=0.7	0.70	0.698 (sd 0.021)	**PASS**	Measures long-term memory: 0.5 = none, above 0.5 = trends persist, below 0.5 = it bounces back
DFA scaling exponent: fGn H=0.3	0.30	0.304 (sd 0.013)	**PASS**	Measures long-term memory: 0.5 = none, above 0.5 = trends persist, below 0.5 = it bounces back
DFA scaling exponent: random walk	1.50	1.501 (sd 0.047)	**PASS**	Measures long-term memory: 0.5 = none, above 0.5 = trends persist, below 0.5 = it bounces back
ACF of AR(1) with phi=0.8: lag-1 value	0.800	0.804	**PASS**	It measures how strongly each sample depends on the previous one
Ljung-Box whiteness test: AR(1) rejected / white noise accepted (25 trials each)	AR rejected 100%, white accepted ~95%	AR rejected 100%, white accepted 100%	**PASS**	Tells random noise from signals that have memory
Sample entropy vs independent reference implementation (white noise)	same value as reference	SYNAPSE 2.195 vs reference 2.193	**PASS**	Its 'how unpredictable is this signal' number matches a trusted calculation
Permutation entropy runs on a current NumPy install (2.x)	returns a number	CRASH on NumPy 2.5.4: module 'numpy' has no attribute 'math'	**FAIL**	A one-word bug (np.math was removed in NumPy 2.0) makes this function crash on a fresh install
Permutation entropy maths (after the 1-line fix): white noise vs a clean sine	noise ~1.0 (max), sine much lower	noise 1.000, sine 0.512	**PASS**	Random data scores near the maximum, orderly data scores low - the formula itself is right
Lyapunov exponent: logistic map r=3.9 (chaotic)	0.495 per step (exact, from the map's own derivative)	0.500 (sd 0.014)	**PASS**	Measures how fast tiny differences blow up - the signature of chaos
Lyapunov exponent: logistic map r=3.7 (chaotic)	0.356 per step (exact, from the map's own derivative)	0.365 (sd 0.007)	**PASS**	Measures how fast tiny differences blow up - the signature of chaos
Lyapunov exponent: Henon map	0.419 per step (published value)	0.407 (sd 0.013)	**PASS**	Same test on a second famous chaotic system
Lyapunov: perfectly clean sine wave (no noise at all)	'not chaotic'	raises an error (insufficient divergence points) instead of answering	**FAIL**	On a perfect periodic signal the engine stops with an error; it should just say 'not chaotic'
Lyapunov: sine wave with 0.1% noise	not chaotic	lambda 0.006, R2 0.02, chaotic=False	**PASS**	A regular wave with tiny noise is correctly not called chaotic
Lyapunov: white noise	not chaotic	lambda 0.034, R2 0.33, chaotic=False	**PASS**	Pure randomness is correctly not called chaos
Higuchi fractal dimension: white noise / Brownian motion / sine	2.00 / 1.50 / ~1.00	2.00 / 1.50 / 1.03	**PASS**	Roughness score: jagged noise ~2, wandering ~1.5, smooth wave ~1
RQA determinism vs independent reference (400 samples: noise / sine / Lorenz)	equal to reference (<=1 point)	noise: 22.6% vs 22.6%; sine: 95.2% vs 95.2%; Lorenz: 93.2% vs 93.2%	**PASS**	Exact match with the reference at 400 samples
RQA determinism vs independent reference (2000 samples: noise / sine / Lorenz)	equal to reference (<=1 point)	noise: 0.0% vs 18.1%; sine: 100.0% vs 95.4%; Lorenz: 44.6% vs 93.9%	**FAIL**	Determinism = how repeatable the pattern is. Exact match up to 400 samples; above that the code thins the data to 400 points and under-reports it
Granger causality: X drives Y (25 trials)	X->Y found 100%; Y->X found <=5%	X->Y 100%; false reverse 8%	**PASS**	It finds who is influencing whom - and the direction
Granger causality: false-link rate on two unrelated noises (300 trials, alpha=5%)	5% (the stated error rate)	10.0% false links	**FAIL**	Unrelated signals get linked about twice as often as promised, because the lag is chosen using the other signal
Transfer entropy: correct direction of information flow (12 trials)	X->Y larger than Y->X in 100%	12/12 correct; mean net flow 0.457 bits	**PASS**	A second, independent way of finding which signal leads
Mutual information (KSG) on correlated Gaussians: rho = 0, .5, .8, .95	exact: 0.000, 0.208, 0.737, 1.679 bits	0.002, 0.213, 0.757, 1.716 bits	**PASS**	How much knowing one signal tells you about the other - compared to the exact textbook formula
Cross-correlation: recover a 25-sample delay (0.125 s)	25 samples	mean error 0.00 samples	**PASS**	It measures the time delay between two signals
Coherence: shared 30 Hz tone buried in independent noise	peak at 30 Hz	mean peak 29.69 Hz	**PASS**	It spots the frequency two signals have in common
PCA explained variance for known covariance (eigenvalues 7:2:1)	70% / 20% / 10%	70.2% / 20.0% / 9.9%	**PASS**	It splits the data into the correct strengths of hidden directions

Test	Known answer	SYNAPSE result	Pass?	What it means
Benjamini-Hochberg: realised false discovery rate at q=0.05 (150 simulations)	<= 5% (theory: 4.5%)	4.85% false discoveries; power 56%	**PASS**	Among the things it calls 'discoveries', no more than 5% should be false
Benjamini-Hochberg: identical decisions to statsmodels reference	100% identical	identical in all 150 runs	**PASS**	Matches the trusted professional library exactly
percentile bootstrap 95% CI coverage (300 skewed datasets, n=60)	95% (92-96% typical)	94.3%	**PASS**	A 95% interval should contain the true value about 95% of the time
BCa bootstrap 95% CI coverage (300 skewed datasets, n=60)	95% (92-96% typical)	93.7%	**PASS**	A 95% interval should contain the true value about 95% of the time
FT surrogates keep the power spectrum exactly	relative error ~ 0	3.46e-16	**PASS**	Shuffled copies keep the same frequencies - only timing is scrambled
IAAFT surrogates keep the value distribution exactly and spectrum closely	distribution error ~ 0, spectrum corr ~ 1	dist err 0.00e+00; spectrum corr 1.0000	**PASS**	Fair 'null' copies for testing whether a pattern is real

B. In-browser math (what the web pages compute) - 18 of 28 passed

Test	Known answer	SYNAPSE result	Pass?	What it means
Welch PSD: find a single hidden tone (25 random freqs, 6-85 Hz)	within 1 bin (<= 0.78 Hz)	25/25 within 1 bin; mean error 0.175 Hz; worst 0.367 Hz	**PASS**	It finds the right frequency even when noise is as strong as the signal
Welch PSD: total power equals the signal's actual variance	same as variance (<=3% err)	mean err 0.64%; worst 1.29%	**PASS**	The energy it reports matches the true energy
Multitaper F-test: recover 3 tones (8.3, 21.7, 44 Hz) in noise	3 of 3 tones every trial	recall 23/45 = 51%	**FAIL**	It finds all three hidden frequencies
Multitaper F-test: false tones on pure noise (30 runs)	<= 1 false line per run	257.00 false lines per run	**FAIL**	On signals with no pattern it should almost never claim a tone
Lomb-Scargle: tone in unevenly spaced samples (20 trials)	error < 0.05 Hz	20/20 correct; mean error 0.0128 Hz; worst 0.024 Hz	**PASS**	It works on data with gaps / irregular timing
Moments of exponential(1): skewness 2, kurtosis	skew 2.00, kurtosis 6 (excess) or 9	skew 2.05, kurtosis 6.16	**PASS**	Basic shape numbers match textbook values
z-outliers (3-sigma rule): find 5 planted 7-sigma spikes (20 trials)	recall 100%	recall 100%, precision 66% (53 extra flags in 40,000 samples)	**PASS**	It finds every real spike, but the fixed 3-sigma cutoff also flags ~3 normal samples per 1000 (53 extra in 40,000). On screen it is labelled 'Rosner ESD', which it is not
z-outliers (3-sigma rule): flags on spike-free noise	2.7 per 1000 (theory for a 3-sigma rule)	377 flags in 120000 samples (3.1 per 1000)	**PASS**	Behaves exactly as a 3-sigma rule should - but that means roughly 3 'outliers' per 1000 samples even in clean noise
DFA scaling exponent: white	0.50	0.504 (sd 0.013)	**PASS**	Long-term memory: 0.5 none, above 0.5 trends persist
DFA scaling exponent: fgn07	0.70	0.702 (sd 0.015)	**PASS**	Long-term memory: 0.5 none, above 0.5 trends persist
DFA scaling exponent: fgn03	0.30	0.307 (sd 0.012)	**PASS**	Long-term memory: 0.5 none, above 0.5 trends persist
DFA scaling exponent: randomwalk	1.50	1.470 (sd 0.052)	**PASS**	Long-term memory: 0.5 none, above 0.5 trends persist
Permutation entropy: white noise vs sine (scale 1)	noise ~1.0, sine much lower	noise 0.998, sine 0.510	**PASS**	Random data scores near 1, orderly data low
Sample entropy of white noise (scale 1)	~2.2 (independent reference 2.19)	2.071	**PASS**	Its unpredictability number matches a trusted calculation
PELT/CUSUM: locate a +1.5 sigma mean shift at sample 1000 (15 trials)	found within 25 samples; no false alarms	found 13/15; median error 9 samples; false alarms on no-change data 0/15	**PASS**	It finds when behaviour changed without inventing changes
Granger causality: X drives Y (25 trials)	X->Y found 100%; Y->X <= ~5%	X->Y 44%; false reverse 0%	**FAIL**	It finds who influences whom
Granger causality: false-link rate on unrelated noises (100 trials, alpha=5%)	5%	0% false links	**PASS**	Unrelated signals should rarely be linked
Transfer entropy: correct direction (12 trials)	X->Y larger in 100%	12/12 correct	**PASS**	A second way of finding which signal leads
Coherence: shared 30 Hz tone in independent noise	peak at 30 Hz	mean peak 48.91 Hz (range 0.0-98.4)	**FAIL**	It spots the frequency two signals share
Cross-correlation: recover a 25-sample (0.125 s) delay	25 samples	mean error 0.00 samples	**PASS**	It measures the time delay between two signals
PCA explained variance for known covariance (7:2:1)	70% / 20% / 10%	76.0% / 24.0% / NaN%	**FAIL**	Correct strength of the hidden directions
Surrogate spectrum test: frequencies falsely rejected on pure noise (15 runs)	<= 5% of 257 bins (12.8 per run) allowed	3.6 per run	**PASS**	False rejections stay within the promised 5% error rate
MUSIC super-resolution: find 3 tones (8.3, 21.7, 44 Hz)	3 of 3	mean 0.00 of 3 found	**FAIL**	A high-resolution frequency finder
Cepstrum pitch of a harmonic series (fundamental 12 Hz)	12.0 Hz	mean 11.87 Hz (sd 0.21)	**PASS**	Finds the base pitch behind a stack of harmonics
Matrix profile: locate a planted odd segment in a sine (first 600 samples)	within 30 samples	0/4 located	**FAIL**	Finds the one stretch that does not repeat

Test	Known answer	SYNAPSE result	Pass?	What it means
Findings page on a hidden signal (tones 8.3/21.7/44 Hz + spike + shift, 10 trials)	all 3 tones, the spike and the shift	tones 19/30, spike 9/10, regime shift 8/10	**FAIL**	Does the headline "discoveries" list contain what was really planted?
Findings page on PURE NOISE (30 runs) - how many "validated" discoveries appear	0 (there is nothing to find)	259.0 validated findings per run; 30/30 runs report at least one (tones 7710, shocks 30, regime shifts 0)	**FAIL**	A trustworthy tool should report nothing when there is nothing there
Multitaper F-test: same pure noise, different amplitude (x0.1, x1, x10, x100)	same answer at every amplitude	lines flagged: 257, 257, 48, 0 of 257 bins	**FAIL**	A real detection test must not change when you only turn the volume up or down. This one does, so it is not a true F-test

4. Bugs found, where they are, and the fix

#	What is wrong	Where	How to fix
1	UI 'Run analysis' result cards use typed-in numbers (ADF -8.45, Granger F 14.8, IAAFT 18.4, coherence 0.92, noise exponent 0.08, status 'supported') and one result depends on the dataset NAME (benchmark_D).	apps/web/src/context/SignalContext.tsx (~lines 520-750)	Call the Python API (/v1/experiments) and show its real numbers, or label the cards 'demo values'.
2	Web page never uses the Python engines; the green 'API OK' only means /health answered.	SignalContext.tsx line ~366	Wire runAnalysis to POST /v1/datasets/upload and /v1/experiments.
3	Paper Workbench shows fixed 'DETERMINISTIC DET 81.2%, lambda +0.14' and 'RQA determinism 84.6%' for every file; dominant mode always 'VALIDATED'.	apps/web/src/views/PaperWorkbench.tsx (~line 340-400)	Compute from data (the Python RQA/Lyapunov already do) or show only when a test passes.
4	In-browser 'multitaper F-test' is not an F-test: it divides power by a power-squared, so the answer changes with signal amplitude (257/257 bins 'lines' at one scale, 0 at another); p-values are made up.	apps/web/src/utils/signalMath.ts ~lines 345-360	Use real DPSS tapers and the Thomson F statistic with a proper threshold.
5	In-browser coherence is a formula with a cosine wobble, not real coherence. Granger uses fixed coefficients (0.4/0.3/0.5) and sets reverse F = 0.25 x forward. PCA returns fixed eigenvalues (68.5/24.6/6.9% or 76/24%).	signalMath.ts: computeCoherenceAndXCorr, computeGrangerCausality, computePCA	Port the Python versions (coherence.py, granger.py, pca.py), which pass their tests.
6	MUSIC returns no peaks even for a clean tone; matrix profile misses a planted anomaly (reports index 22, truth 300).	signalMath.ts: computeMUSIC, computeMatrixProfile	Debug / replace; hide until working.
7	Python BOCPD never detects anything: its change probability is always exactly 1/lambda.	packages/synapse_anomaly/bocpd.py	Use P(run length < k) or the drop of the most likely run length instead of P(r=0).
8	Python permutation entropy crashes on NumPy 2.x (np.math was removed).	packages/synapse_structure/entropy.py line 97	Replace np.math.factorial with math.factorial.
9	Python multitaper lists ~20 false tone lines per noise run (every frequency tested at alpha 0.01 with no correction).	packages/synapse_spectra/multitaper.py	Apply Bonferroni (1/N) or the repo's own Benjamini-Hochberg to the line list.
10	Python Granger gives false links 10% of the time instead of 5% (lag picked using the model that already contains the other signal).	packages/synapse_relation/granger.py	Pick the lag from the restricted (own-past) model, or fix the lag.
11	Python RQA thins the data to 400 points, so determinism is wrong above 400 samples (Lorenz 44.6% vs 93.9%; sine RR 12.3% vs 5.5%). At <=400 samples it matches an independent copy exactly.	packages/synapse_structure/rqa.py (max_pts = 400)	Remove the subsampling or use windows.
12	Python Lyapunov raises an error on a perfectly clean periodic signal instead of returning 'not chaotic'.	packages/synapse_structure/lyapunov.py	Catch the no-divergence case and return not chaotic.
13	Hidden-signal batch of the full Python pipeline and live arXiv search were not scored (time / sandbox has no arXiv access).	-	Re-run on your machine.

Notes on fairness: I first judged CUSUM too harshly (alarm threshold h=5 is naive for 2,000 samples) and a Lyapunov/logistic test used start values that are exact fixed points; both were my mistakes and were corrected. Lorenz Lyapunov (1.85 vs 0.906) overshoots, but my own independent copy of the same algorithm gives 1.82, so that is a method/settings limit, not a coding error. Your own 33 pytest tests all pass.